Project Design Phase
Problem – Solution Fit Template

Date	15 February 2026
Team ID	LTVIP2026TMIDS54574
Project Name	Smart Sorting Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

1. Customer Segment(s) (CS)

Parameter	**Description**
Target Customers	Farmers, Supermarket owners, Food processing industries, Vegetable vendors, Warehouse managers
Secondary Customers	Smart agriculture system developers, Quality control inspectors
Customer Characteristics	Handle large quantities of fruits and vegetables, need quality assurance, want to reduce food wastage, require fast and accurate detection systems

2. Jobs-To-Be-Done / Problems (JBP)

Parameter	**Description**
Functional Jobs	Identify whether fruits and vegetables are fresh or rotten
Operational Jobs	Sort large quantities of produce quickly and accurately
Quality Control Jobs	Ensure only fresh produce reaches customers

Key Problems	Manual sorting is slow, error-prone, inconsistent, and inefficient
Desired Outcome	Automated, fast, and accurate detection system

3. Triggers (TR)

Trigger Type	Description
Operational Trigger	Increase in volume of fruits and vegetables to inspect
Financial Trigger	Losses due to spoiled or rotten produce
Technological Trigger	Availability of AI and image recognition technology
Quality Trigger	Need to improve quality standards and customer satisfaction

4. Emotions: Before / After (EM)

Before Using Solution	After Using Solution
Frustration due to manual work	Satisfaction with automated system
Uncertainty in quality detection	Confidence in accurate detection
Stress due to time consumption	Relief due to faster processing
Fear of financial loss	Trust in reliable quality control

5. Available Solutions (AS)

Existing Solution	Limitations
Manual inspection by workers	Slow, inconsistent, human error
Basic sorting machines	Cannot detect internal or early-stage rot accurately
Visual inspection	Subjective and unreliable
No automated AI-based systems	Lack of intelligent detection

6. Customer Constraints (CC)

Constraint	Description
Cost Constraints	Limited budget for advanced systems
Technical Constraints	Lack of AI-based automated tools
Time Constraints	Manual sorting takes too much time
Accuracy Constraints	Human detection is not always reliable

7. Behaviour (BE)

Behaviour Type	Description
Current Behaviour	Customers manually inspect fruits and vegetables
Alternative Behaviour	Hire workers for sorting
Technology Behaviour	Slowly adopting AI and automation tools
Decision Behaviour	Prefer reliable, fast, and cost-effective solutions

8. Channels of Behaviour (CH)

Online Channels

Channel	Usage
Web applications	Upload images and check results
Mobile apps	Monitor produce quality
Cloud platforms	Store and process data

Offline Channels

Channel	Usage
Supermarkets	Quality inspection
Warehouses	Sorting and storage
Food processing units	Automated sorting

9. Problem Root Cause (RC)

Root Cause	Description
Manual Process	Human inspection is slow and inconsistent
Lack of Automation	No intelligent automated detection system
Large Volume	Difficult to inspect large quantities manually

Human Error	Fatigue leads to mistakes

10. Your Solution (SL)

Solution Aspect	**Description**
Solution Name	Smart Rotten Fruit & Vegetable Detection System
Core Technology	Transfer Learning with Deep Learning models
Functionality	Detects fresh or rotten produce using image classification
User Interaction	User uploads image through web application
Output	Displays prediction result instantly
Benefits	Faster detection, higher accuracy, reduced food wastage, improved efficiency